Chapter 23

Climate Risk, ESG, and Sustainability

We now move on to consider environmental risks. These are the risks that we will damage the environment so much that the Earth will become less habitable (or even totally uninhabitable) for future generations. All companies, including financial institutions, are finding that, as a result of societal and other pressures, environmental risks can no longer be ignored. It seems almost certain that these risks will become increasingly important in the years ahead. The biggest environmental risk is climate risk, and this is what the first part of this chapter focuses on.

23.1 Climate Risk

The inhabitants of planet Earth add tens of billions of tons of greenhouse gases to the atmosphere each year. This is the major cause of global warming. When the energy from the sun reaches the Earth, some is absorbed and some is radiated back as heat. Increases in greenhouse gas concentrations in the atmosphere lead to less heat finding its way into space and more being absorbed by the atmosphere and the oceans. Unfortunately, this problem is not naturally self-correcting, as greenhouse gases stay in the atmosphere for hundreds of years.

The most prevalent greenhouse gas is carbon dioxide (CO_2).[1] As is well known, this is emitted by the burning of fossil fuels such as coal, oil, and natural gas. But it also arises from many other activities such as manufacturing cement, concrete, steel, and plastics, and raising animals for food. Achieving a reduction in greenhouse gas emissions is now recognized as essential for the survival of the human race. The Paris Agreement in 2015 had a goal of limiting temperature rises above preindustrial levels during the twenty-first century to 1.5 degrees Celsius. The Glasgow Climate Pact in 2021 reaffirmed this objective and strengthened targets for the year 2030 with a goal of reaching net-zero emissions by the year 2050. It is expected that the progress will be reviewed at further meetings between countries at intervals of about five years.

How successful we will be in achieving the goals set in Paris and Glasgow remains to be seen. Arguably, a temperature rise of less than 1.5 degrees Celsius is optimistic. The actual rise during the twenty-first century could be as high as 4 degrees Celsius or even 6 degrees Celsius. The primary reason for the broad range of estimates is not the physical science but the uncertainty around the global policy response and its success in actually reducing emissions. A number of organizations have developed models and produced climate scenarios. These organizations include the Intergovernmental Panel on Climate Change (IPCC), the Network for Greening the Financial System (NGFS), and the International Energy Agency (IEA). The IPCC calls these scenarios Shared Socioeconomic Pathways (SSPs) and published its sixth assessment report in 2021 (Table 23.1).[2]

The key problem is that humans have to make a trade-off between the emission of greenhouse gas today and the resultant damage to the environment in the future, which will impact their children and grandchildren. There is no easy way to make this trade-off. Table 23.2 shows 2020 estimates of greenhouse gas emissions by sector, produced by

Table 23.1 Shared Socioeconomic Pathways in IPCC Sixth Assessment Report

SSP	**Scenario**	**Estimated Warming, 2041–2060, °C**	**Estimated Warming, 2081–2100, °C**	**Very Likely Range[3] 2081–2100, °C**
SSP1	CO_2 emissions cut to net zero by about 2050	1.6	1.4	1.0–1.8
SSP2	CO_2 emissions cut to net zero by about 2075	1.7	1.8	1.3–2.4
SSP3	CO_2 emissions continue at about current level until 2050 and then fall	2.0	2.7	2.1–3.5
SSP4	CO_2 emissions double by 2100	2.1	3.6	2.8–4.6
SSP5	CO_2 emissions triple by 2075	2.4	4.4	3.3–5.7

[1]Other greenhouse gases that are created by human activities include methane, nitrous oxide, halocarbons, and ozone.

[2]See IPCC Sixth Assessment Report, 2021, www.ipcc.ch/assessment-report/ar6.

[3]This can be interpreted as a 95% confidence interval.

Table 23.2 Greenhouse Gas Emissions by Economic Sector in the United States in 2020

Sector	**Emissions**
Transportation	27%
Electricity	25%
Industry	24%
Commercial and residential	13%
Agriculture	11%

the U.S. Environmental Protection Agency.[4] Eliminating greenhouse gas emissions will require many changes to the way we produce energy, transport ourselves, manufacture materials, and feed ourselves. It will not be easy for developed countries. Furthermore, it may be an unreasonable burden for developing countries, as it is likely to adversely affect their economic progress. China is responsible for the most greenhouse gas emissions (about 10 billion tons per year) followed by the United States (about 5 billion tons per year) and India (about 2.5 billion tons per year). But on a per capita basis, the United States emits about twice as much as China and seven times as much as India.[5]

Climate change is going to be accompanied by rising sea levels as ice melts and sea water expands. It has been estimated that the sea level will rise a few meters for each degree Celsius of global warming.[6] Some CO_2 will be absorbed by oceans, creating ocean acidification that adversely affects ocean ecosystems. Precipitation is expected to increase by about 3% for every degree Celsius of global warming because of evaporation from oceans.[7] However, this will not be spread evenly. Areas that already get a lot of rain will get more while areas that are dry will become drier. As a result, floods can be expected in some parts of the world. Other extreme weather events such as storms, heat waves, wildfires, hurricanes, and tornadoes can also be expected. Indeed, we are already seeing them, and their costs total billions of dollars. In 2021, it is estimated that Hurricane Ida in the United States cost $75 billion, flooding in China cost $30 billion, and flooding in Germany and Belgium cost $22 billion.[8] The impact on human life is liable to be severe. In 2003, a heat wave in Europe caused tens of thousands of deaths. In 2022, parts of the world experienced an even worse heat wave.

[4]See EPA, "Sources of Greenhouse Gas Emissions," www.epa.gov/ghgemissions/sources-greenhouse-gas-emissions.

[5]See www.worldometers.info/co2-emissions/co2-emissions-by-country for statistics on all countries.

[6]See J. Garbe, T. Albrecht, A. Levermann, J. F. Donges, and R. Winkelmann, "The Hysteresis of the Antarctic Ice Sheet," *Nature* 585 (September 2020): 538–544.

[7]See N. Jeevanjee and D. M. Romps, "Mean Precipitation Change from a Deepening Troposphere," *Proceedings of the National Academy of Sciences* 115 (45), (October 2018): 11465–11470.

[8]See Jeff Masters, "The Top 10 Climate Change Events of 2021," Yale Climate Connection (January 11, 2022), https://yaleclimateconnections.org/2022/01/the-top-10-global-weather-and-climate-change-events-of-2021/.

In this chapter, we consider the effect of climate risk on financial institutions. There are many uncertainties. Possibly, technological innovations (e.g., a cost-efficient way of removing CO_2 from the atmosphere and storing it permanently, or a cost-efficient way of capturing CO_2 before it is released into the atmosphere and storing it permanently will happen). But it is also possible that international agreements prove to be ineffective. There could be a ripple effect. If one or two large countries continue to be high emitters, a few other countries might decide not to follow through on their plans to reduce emissions because it will make little overall difference and could disadvantage them economically. This, in turn, could lead to yet more countries abandoning their green initiatives, and so on. This is a concern because there has already been a tendency for large countries to make climate commitments and then not follow through on them.

There have been some encouraging developments. One has been the establishment of the Task Force on Climate-Related Financial Disclosures (TCFD) by the Financial Stability Board. This develops recommendations on information related to climate risks that companies should disclose. Led by Michael Bloomberg, TCFD was supported by 3,000 organizations in 92 countries with a combined market capitalization of $27.2 trillion in 2022. As we discuss later, Mark Carney's Glasgow Financial Alliance for Net Zero (GFANZ) has lined up vast amounts of investment capital that are committed toward transitioning to a low carbon future.

Analysts distinguish *physical risks* from *transition risks*. Physical risks are the costs of extreme climate events such as floods, rising sea levels, hurricanes, and so on. Transition risks are the risks associated with the move to lower emissions. These risks include the introduction of carbon taxes and costs incurred to reduce emissions by changing business practices or discontinuing certain activities altogether. For example, companies may find it necessary to close down plants that produce energy from fossil fuels earlier than planned. Business Snapshot 23.1 provides an example of this. However, the transition to lower emissions may give rise to opportunities for some companies as they find environmentally friendly ways to produce electricity, cement, and other products that currently lead to the production of greenhouse gases.

Other relevant risks related to climate change are those associated with large capital flows that may alter financial markets. GFANZ estimates that transitioning the economy will require additional investments on the order of $100–150 trillion over the next 30 years. Finally, there are the risks of individual countries taking somewhat radical steps to mitigate the effects of climate change. For example, Solar Radiation Management, a form of geo-engineering that increases the ability of the atmosphere to reflect sunlight, is a short-term response that could have a temporary cooling effect that is both noticeable as well as quite affordable (only a few billion dollars).[9] However, the consequences of such actions are very uncertain and would likely affect ocean currents, create droughts, or rainfall events in neighboring countries, as well as having biological impacts. In spite

[9]See, for example, Simon Nicholson, "Solar Radiation Management," Wilson Center, September 30, 2020, https://www.wilsoncenter.org/article/solar-radiation-management.

BUSINESS SNAPSHOT 23.1

Transition Costs in Germany

In July 2020, Germany passed a law requiring that all coal-fired plants be closed by 2038 with some compensation being provided to owners. This will lead to many plants being closed down before the end of their useful lives (typically 30–50 years). For example, a coal power station, Datteln, became operational in 2020 and would have a maximum life of only 18 years if closed as planned. There will be a cost to German taxpayers.

of this, there are early pilot programs underway in an attempt to delay the effects of climate change for the most affected countries.

Governments have introduced some measures to reduce CO_2 emissions. About 25 countries have introduced carbon taxes. Sweden was the first country to do so in 1991 and has increased the tax rate per ton of fossil CO_2 emitted in 1997 and 2007. According to studies that have been carried out, the tax has been instrumental in reducing emissions. In the United States, a carbon tax has been repeatedly proposed but not implemented. Another measure is referred to as *cap and trade*. The amount of CO_2 that a company emits is capped, but companies can trade with each other. If a company does not need to emit its full quota, it can sell the unused part to another company that wants to emit more than its quota. This can lead to emissions being reduced where it is easiest and cheapest to do so. Cap and trade schemes are used in many jurisdictions, including China and some U.S. states.

23.1.1 Insurance Companies

Climate changes are likely to have a big impact on the insurance industry. Ideally, insurance companies would like to take on risks that are uncorrelated with each other. Consider the car insurance portfolio of a large insurance company. The claims made by different policy holders have very little correlation with each other. This means that a "law of large numbers" argument applies and the total payout in a year can be predicted reasonably accurately. We made the point in Chapter 3 that the same is not true for all of the risks the insurance company takes on. A climate-related disaster such a Hurricane Ida leads to many policy holders making claims at the same time. It seems almost certain that these types of natural disasters will occur more frequently in the future.

Insurance companies have over the years developed a great deal of expertise in modeling catastrophic risks. Their models must now recognize that the frequency and severity of these risks are increasing.[10] The actuarial practice of averaging the cost of past risks

[10]See The Geneva Association, "Climate Change Risk Assessment for the Insurance Industry," 2021. https://www.genevaassociation.org/sites/default/files/research-topics-document-type/pdf_public/climate_risk_web_final_250221.pdf.

to estimate the cost of future risks is not valid. Fortunately, the vast majority of property and casualty insurance and reinsurance contracts are for one year. This makes it possible to adjust the pricing, terms and conditions, and product offerings as climate conditions worsen.

Unlike property and casualty companies, life insurance companies do not have the flexibility of changing their premiums as conditions worsen. The annual premiums for whole life policies usually remain fixed for the life of the policy holder. As explained in Chapter 3, life expectancy in the United States has increased by about 20 years over the last 100 years. This trend has benefited life insurance companies because it means that premiums are received for longer and payouts are later than expected. It is possible that temperature increases and other disruptions will lead to a reversal of the life expectancy trends. Whole life policies might then become less profitable.

There can be no doubt that climate risks will make the management of insurance companies more challenging. Insurance companies must devote resources to understanding climate trends and quantifying their effect on risks they are underwriting. They need to convince regulators that they are handling climate risks in a responsible way. It is clearly in their interests and in the interests of society that policy holders be educated on ways of managing climate risks.[11] But an unfortunate reality is that insurance companies are becoming increasingly unwilling to provide insurance for natural disasters, such as floods, in areas they consider to be high risk. An important question for society is likely to be who bears the cost of damage to properties from natural disasters. Is it the property owners, private insurance companies, a government-sponsored insurance company, or disaster relief programs?

23.1.2 Banks

Banks are being proactive in trying to quantify the effects of climate change. The Partnership for Carbon Accounting Financials (PCAF) is a forum where banks exchange ideas on the best way to measure and address carbon emissions. The risks for banks arising from climate changes are also considered by their regulators. The European Central Bank in 2022 launched a supervisory climate stress test to assess how prepared banks are for dealing with financial and economic shocks arising from climate risk.[12] In a 2021 document, the Basel Committee on Banking Supervision argued that the impact of climate changes can be understood by considering their effect on traditional risk categories. Table 23.3 summarizes the Basel Committee's conclusions.[13]

[11]For example, there are a number of simple steps home owners can take to avoid basement flooding. See https://www.intactcentreclimateadaptation.ca/wp-content/uploads/2021/03/3-Steps-to-Home-Flood-Protection_March-2021_Space-for-Partner-Logo.pdf?utm_source=kitchenertoday.com™utm_campaign=kitchenertoday.com&utm_medium=referral.

[12]See European Central Bank, "SCU Banking Supervision Launces 2022 Climate Risk Stress Test," Press release, January 27, 2022, bankingsupervision.europa.eu-ECB Banking Supervision launches 2022 climate risk stress test.pdf.

[13]See Basel Committee on Banking Supervision, "Climate-related risk drivers and their transmission channels." Bank for International Settlements, April 2021, www.bis.org/bcbs/publ/d517.pdf.

Table 23.3 Potential Impact of Climate Risks on Banks

Risk	Potential Effects of Physical and Transition Climate Risks
Credit risk	Climate risk may reduce borrowers' ability to service debt or the value of collateral in the event of a default.
Market risk	Prices of financial assets may suffer as the impact of climate risk becomes more apparent.
Liquidity risk	A bank's access to stable sources of funding such as retail deposits could be reduced as market conditions change.
Operational risk	There are likely to be legal and regulatory compliance risks associated with corporate clients and investments.
Reputational risk	Reputational risk may increase as a result of market or consumer sentiment.

Climate changes affect the risks associated with a bank's mortgage portfolio. As already pointed out, premiums can be changed each year in property and casualty insurance. By contrast, a mortgage entails a commitment over several years. For example, a mortgage in the United States typically lasts 15 or 30 years.

The occurrence of climate-related events, or the market at some future time becoming more aware of the possibility of these events, can affect real estate prices and therefore the value of mortgage collateral. As we saw during the Global Financial Crisis (see Chapter 7), when the value of a house drops below the amount outstanding on the mortgage, the homeowner may have an incentive to default even if he or she has the financial resources to continue to service the mortgage. A default leads to the bank taking possession of the house and having to sell it in a difficult market where other similar properties are being sold for similar reasons. In some jurisdictions, the bank has no recourse to other assets of the borrower.

It is not difficult to imagine climate risks causing similar problems to those created by the Global Financial Crisis. The possibility of rising sea levels, for example, could become an increasing concern in certain areas, reducing house prices and leading owners to bail out by defaulting on mortgages. When valuing a house for the purposes of determining the amount that can be safely lent, it is clearly important that banks take account of the potential impact of adverse climate events.

The evidence is that severe climate-related events can reduce corporate profitability and therefore increase credit risk to their lenders.[14] Supply chains can be disrupted, worker productivity can be adversely affected, and consumer purchasing patterns can change. The COVID-19 pandemic had an adverse effect on the profitability of some businesses. The impact of severe climate-related events in certain parts of the world could be similar.

There are other factors that banks and their corporate clients need to consider. Climate changes are likely to lead to mass migrations as individuals seek to leave

[14]In 2019, Pacific, Gas & Electric (PG&E) a California company, filed for bankruptcy because it was blamed for wildfires and lawsuits resulted. This has been dubbed the first climate-related bankruptcy.

unbearably hot climates, areas that have been flooded, areas where clean water is no longer available, or areas that have become unsuitable for farming. Migration can happen within countries or between countries. This can be expected to lead to widespread economic disruption and create difficult problems for governments to handle. The impact on agriculture will be mixed. In some northerly regions, crop yields can be expected to increase as a result of warmer climates. In other areas, crops that have been grown for centuries will no longer be viable because the land is too dry. Rearing healthy animals for food will be more difficult in many parts of the world. Food shortages are likely. Pests that threaten crops grown in different regions will also migrate with the changing climate, and this can cause crop failures and periods of low yields as farmers learn to deal with these problems. Farmers who are desperate for income might require increasing amounts of fertilizer to deal with difficulties caused by climate change. This might accentuate difficulties, as fertilizer production is a major cause of greenhouse gas emission.

Sovereign debt is likely to be affected by climate change. As companies are adversely affected by climate risks, they become less profitable. As a result, tax revenue declines and the demand for borrowing by countries rises. Unfortunately, less-developed countries will tend to be most affected by climate changes and climate-related disasters. Furthermore, these are also countries that need capital to enable them to move away from high-emitting sources of energy (coal and wood burning) and high-emitting agricultural practices. This will present a challenge for banks and the governments of developed countries and may lead to an increase in global tensions.

The transition to a low-carbon economy means that there will be winners and losers. Electric vehicle producers can be expected to do well, as can companies that are concerned with methods of producing energy without fossil fuels (e.g., by using wind or the sun). Oil companies and coal companies can expect difficult times ahead. Royal Dutch Shell was sued by Millieudefensie (Friends of the Earth Netherlands), an environmentalist group, and ordered by a court in the Hague in 2021 to reduce its carbon emissions to 45% of 2019 levels by 2030. (Shell is appealing the ruling.) Such lawsuits may become more common. Some observers consider carbon emitters to be analogous to tobacco companies, which have in recent years been on the losing side of lawsuits concerning the impact of cigarettes on a smoker's health.

Banks are concerned about reputational risk. Suppose that a bank considers a loan to a carbon emitter to have low credit risk. It might still refuse to make the loan because it could lead to adverse publicity and environmentally conscious customers closing their accounts. Indeed, banks are coming under pressure from many different organizations to take climate risk into account in their lending decisions. Goldman Sachs, for example, takes reputational risk seriously. In its TCFD report, it states that involvement in industries associated with climate change may lead to "reduced client and employee loyalty, investor divestment and impacts to client activity." Business Snapshot 23.2 provides another illustration of a financial institution wanting to avoid reputational risks.

BUSINESS SNAPSHOT 23.2

Reputational Risk and HSBC

An interesting story concerns a *Financial Times* event in May 2022. Stuart Kirk, Global Head of Responsible Investments at the bank HSBC, delivered a speech titled "Why investors need not worry about climate risk." He first argued that human beings have proved throughout the ages that they are adaptable and manage changes well. He then argued that if an average growth rate in the economy of 2% or more continues until the year 2100, the negative effects of climate change will be manageable because they will be small compared with the gains experienced from economic growth.

Whether you agree with him or not, Mr. Kirk is entitled to his opinion. (News reports after the presentation suggested that many asset managers, and some politicians, would agree with him.) However, HSBC quickly distanced itself from the presentation and suspended him. (This is in spite of the fact that it approved the title of the presentation in advance.) Why did HSBC do this? The reason concerns reputational risk. Financial institutions want to be (and be seen to be) environmentally responsible. It is not in their interest to incur the displeasure of environmental groups and lose business as a result.

In considering arguments such as those of Mr. Kirk, we should keep in mind that climate risks are not like other risks to which the human race has adapted. They are risks that affect the whole planet and its ability to support human life as we know it. It is encouraging that many banks, and their regulators, are trying to act responsibly. The speed and scale of the human response that is required to address climate change means that financial markets and their participants will play an important role in reducing emissions.

23.1.3 Institutional Investors

Institutional investors are increasingly taking climate risk into account in forming portfolios. The $240 billion California State Teachers' Retirement System was a pioneer. It formed a Green Initiative Task Force in 2007 to identify, analyze, and propose investment opportunities addressing climate change. The company is active in promoting environmentally friendly behavior (sometimes referred to as corporate social responsibility, CSR) at shareholder meetings.

Companies have an incentive to demonstrate that they are reducing emissions in line with targets. If institutional investors refuse to invest in a certain company because its climate-related disclosures are unsatisfactory, the company is likely to suffer. To put this another way, investors have a great deal of leverage over the companies they invest in and can play a key role in persuading them to reduce greenhouse gas emissions.

Green bonds are an interesting innovation. In 2007, the European Investment Bank issued a "climate awareness bond," and in 2008 the World Bank used the term *green bond* to describe bonds where the proceeds are used for climate-friendly investments. Green bonds are now issued by both the public and private sector. Investment banks handle the underwriting and an opinion on the environmental credentials of the issuer is often given by an external agency. Other related innovations are green loans and sustainability-linked bonds and loans.

We have already mentioned TCFD, the Task Force on Climate-Related Disclosures, which was formed in 2015 by the Financial Stability Board to develop recommendations on climate-related disclosures. Its recommendations involve four themes: governance, strategy, risk management, and metrics. It recommends that companies disclose:

- How climate risks and opportunities are managed
- The current and potential future impact of climate risks
- The process used to identify the risks
- The metrics used to quantify greenhouse gas emissions

The recommendations have received widespread support. As of October 6, 2021, the Task Force had over 2,600 supporters globally, including 1,069 financial institutions responsible for $194 trillion of assets. Jurisdictions such as Brazil, the European Union, Hong Kong, Japan, New Zealand, Singapore, Switzerland, and the United Kingdom have introduced or announced their intention to introduce mandatory disclosure requirements that are aligned with TCFD.[15]

One issue is that a number of different organizations are trying to increase transparency and measurability of climate-related financial reporting. Similar to the financial reporting industry, coordination is vital. For financial reports, large umbrella organizations like IFRS and FASB led the effort toward global coordination of accounting standards to ensure that investors can understand and rely on corporate reports and avoid falling prey to financial reports that have misleading or fraudulent content. Similarly, organizations like TCFD are working very hard to get sustainability-related reports to be credible and comparable. However, these efforts are still a work in progress, and the attendant uncertainty presents yet another risk to investors.

In June 2021, 457 investors with over $41 trillion in assets under management signed the 2021 Global Investor Statement to Governments on the Climate Crisis. Others have signed since. The Statement was a call for governments to undertake five actions:

1. Strengthen nationally determined policies to limit warming to 1.5 degrees Celsius.
2. Commit to zero emissions by 2050.
3. Strengthen carbon pricing, phase out fossil fuel subsidies and coal-based power, and avoid new carbon-intensive infrastructure.

[15]See The Financial Stability Board, Task Force on Climate-Related Disclosures, Bank for International Settlements, October 2021, https://assets.bbhub.io/company/sites/60/2021/07/2021-TCFD-Status_Report.pdf.

4. Ensure that COVID-19 economic recovery plans are consistent with the transition to net zero emissions.
5. Commit to introducing mandatory climate risk disclosures.

Mark Carney, former governor of the Bank of Canada and the Bank of England, has led an effort to ensure that financial institutions account for and disclose the climate risks in their lending and investments. As United Nations special envoy for Climate Action and Finance, he chaired the Glasgow Financial Alliance for Net Zero (GFANZ), which was formed ahead of the 2021 Glasgow summit. This is a global coalition of leading financial institutions committed to accelerating the decarbonization of the economy.

It should be clear from this that there are many initiatives aimed at using investors to persuade companies to reduce, and eventually eliminate, greenhouse gas emissions. Some asset managers such as BlackRock have embraced the net zero objective enthusiastically. Others such as Berkshire Hathaway, a conglomerate headed by Warren Buffett, have been reluctant to disclose climate risk metrics, but are coming under increasing pressure to do so. Unfortunately, the pressure is not all one way. In July 2022, Florida's governor proposed legislation that would ban the state's pension funds from selecting investments based on environmental factors. In August 2022, attorney generals from 19 U.S. states wrote to BlackRock CEO Larry Fink, asking him to focus on profits and not take environmental issues into account in state pension investments.

To summarize: companies are under pressure (for financial, societal, and legal reasons) to disclose their current emissions and develop plans for reducing them. As just mentioned, not all politicians have been supportive of this. But, in spite of that, there is an increasing focus on emissions, and it is to be hoped that this will result in climate change proving to be manageable.

23.2 ESG

ESG stands for environmental, social, and governance. Whereas it used to be considered that it was the job of a company to maximize the profits for shareholders, it is now recognized that the profit objective must be combined with environmental and social responsibility.[16]

ESG is often considered to be synonymous with climate risk, but ESG has somewhat broader concerns than climate. The "S" covers social issues such as diversity, equity, inclusion, and worker well-being. The metrics to measure social factors are not as well developed as those for environmental factors, but it is likely that companies will be required to disclose more information in this area in the future. How does a company manage discrimination and harassment in the workplace? What steps is it taking to encourage gender, racial, and ethnic diversity in its management of human capital?

[16] A famous (now dated) quote from Nobel Prize-winning economist, Milton Friedman in 1970 is "There is only one social responsibility of business: to use its resources and engage in activities designed to increase its profits."

Some large companies in the United States are recognizing that the social landscape is changing and are changing their internal practices. The Securities and Exchange Commission (SEC) has approved a proposal by the stock exchange Nasdaq to require listed companies to have one director who is female and another who is black, Hispanic, Native American, LGBTQ+, or part of another underrepresented group (or to explain why they do not have this level of diversity). In 2020, Goldman Sachs announced that it would stop underwriting IPOs for companies in the United States and Europe that do not have diverse boards.

The "G" in ESG is concerned with companies using accurate and transparent accounting methods and having governance mechanisms in place to ensure policies and practices are actually followed. A company should be accountable to shareholders and avoid conflicts of interest in their choice of board members and senior executives. It should not make political contributions in order to obtain preferential treatment. It should also not engage in illegal activities such as bribing foreign officials to obtain contracts. The CEO and chairperson of the board should be different individuals. Measures of good governance have existed for some time. For example, Daniel Kaufmann from the Brookings Institute and Aart Kray from the World Bank have produced Worldwide Governance Indicators (WGI) for many years.

One of the difficulties in this area, similarly to the area of climate accounting, is that there are many different providers of ESG data. Not surprisingly, these ESG reports from different data providers often disagree with each other. Since there is currently no regulation concerning ESG reporting standards, and since ESG data providers are competing for business by creating reports that users will pay for, there are many ways for firms to achieve a good ESG rating. Firms can, effectively, shop around for a good ESG score. This lack of standardization and transparency represents a source of risk for investors.

23.3 Sustainability

Like ESG, sustainability includes climate risk, but covers other issues. In December 1983, the secretary-general of the United Nations, Javier Pérez de Cuéller, asked the former prime minister of Norway, Gro Harlem Brundtland, to create an organization that became known as the Brundtland Commission. Its task was to consider environmental strategies for sustainable development. The Brundtland report's definition of sustainable development has been widely used:

> Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their needs.

The production of greenhouse gases is an important example of nonsustainable development, but there are others. The way we dispose of waste is an important consideration. The pollution of oceans with plastic waste, manufactured chemicals, petroleum waste, agricultural runoff, sewage, and more is a particular concern. It kills fish and exposes humans to toxins because they eat contaminated seafood. Another example of nonsustainable development is the contamination of soil and water sources from pesticides, fertilizers, petroleum products, and so on.

The Sustainability Accounting Standards Board (SASB) was founded in 2011 as an independent nonprofit organization to provide sustainability metrics. Another similar initiative is the Global Reporting Initiative, which was founded in 1997 following the Exxon *Valdez* oil spill.

There are considered to be three pillars of sustainability:

- Environmental sustainability
- Economic sustainability
- Social sustainability

We have discussed environmental sustainability. Economic sustainability refers to practices that support the long-term development of a company or nation while also protecting environmental, social, and cultural elements. Social sustainability occurs when today's actions support the capacity of current and future generations to create healthy and livable communities. Sustainability requires that we consume natural resources at a rate where they are able to replenish themselves. Overfishing or deforestation are examples of activities that are not sustainable.

A number of funds with the explicit objective of investing in companies with good ESG or sustainability practices have been developed. They are designed to appeal to retail investors who want to use their investment dollars to make the world a better place. An interesting question is: Do ESG/sustainable funds outperform other funds? Although the demand for these funds has increased, their performance appears to be no better than that of other funds.[17] Unfortunately, some organizations produce reports with questionable research practices that try to convince their customers that "doing well by doing good" is easy and will lead to high returns with low risks. Finance theory suggests that high returns tend to be accompanied by high risks. In any event, earning high returns should not be a primary goal for investing in sustainability-motivated funds.

23.4 Greenwashing

Companies in many industries are keen to present themselves as ecologically responsible. Consider two similar products, one advertised as environmentally friendly and the other promoted in a different way. Many consumers would prefer to buy the first product. This has led to some misleading claims known as *greenwashing*. Here are some

[17]See S. M. Hartzmark and A. B. Sussman, "Do Investors Value Sustainability? A Natural Experiment Examining Ranking and Fund Flows," *Journal of Finance* 74 (6) (December 2019): 2789–2837.

examples of greenwashing from Akepa, an organization that works with companies to create sustainable and ethical brands:[18]

- Innocent drinks, owned by Coca-Cola, suggested in its TV advertising that its plastic products were environmentally friendly when in fact they could not be recycled.
- Keurig made misleading recycling claims in its packaging about plastic pods. It was fined $3 million and ordered to change what it says.
- IKEA has generally good sustainability credentials, but ran into problems because its beechwood chairs were using illegally sourced wood.
- Ryanair claimed, with no evidence, that it was Europe's "lowest emissions airline."

Unfortunately, financial institutions do not have a perfect record as far as greenwashing is concerned. In 2022, Deutsche Bank's asset management subsidiary, DWS, was found guilty of exaggerating the green credentials of the investments it sold. To quote Magdalena Senn of the German consumer advocacy group Finanzwende, "Greenwashing is not a trivial offense." The CEO of DWS was forced to resign.

Summary

Risk management groups now actively consider environmental risks. At most large organizations, there is a senior executive responsible for these risks. As the extent of the risks to our planet become increasingly apparent, environmental risks are likely to become an increasingly important consideration for many companies and countries. The Global Association of Risk Professionals (GARP) now offers a Sustainability and Climate Risk (SCR) qualification in addition to its well-known FRM. The CFA Institute offers a Certificate in ESG Investing in addition to the CFA charter.

Further Reading

Garbe, J., T. Albrecht, A. Levermann, J. F. Donges, and R. Winkelmann, "The hysteresis of the Antarctic Ice Sheet." *Nature* 585 (September 2020): 538–544.

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[18]See "Greenwashing: 10 Recent Stand-Out Examples," Akepa, July 23, 2021, https://thesustainableagency.com/blog/greenwashing-examples.

Practice Questions and Problems (Answers at End of Book)

23.1 Why does the burning of fossil fuels lead to temperature rises?
23.2 What is the goal that was agreed to in Paris in 2015?
23.3 What are SSPs?
23.4 What is the difference between a carbon tax and a cap and trade system?
23.5 What is the difference between physical risks and transition risks?
23.6 What are the risks of climate change to insurance companies?
23.7 What does ESG stand for?
23.8 What is greenwashing?

Further Questions

23.9 How can pension plans and mutual funds help solve the global warming problem?
23.10 Give examples of activities covered by ESG and sustainability that are not climate risk issues.